PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE ONLY), P-CHANNEL
SILICON TYPES 2N7422, 2N7422U, 2N7423 AND 2N7423U
JANTXVR AND F AND JANSR AND F

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a P-channel, enhancement-mode, MOSFET, radiation hardened (total dose only), power transistor. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum rating (E_{AS}) and maximum avalanche current (I_{AS}).

1.2 Physical dimensions. See figure 1, (TO-254AA), and figure 2, (surface mount).

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type 4/	P_T 1/ T_C = +25°C	P_T 1/ T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C =+25°C	I_{D2} 2/ T_C = +100°C	I_S	I_{DM} 3/	T_J and T_{STG}	V_{ISO} 70,000 ft. altitude
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7422	150	4.0	-100	-100	±20	-22.0	-14.0	-22.0	-88	-55 to +150	N/A
2N7423	150	4.0	-200	-200	±20	-14.0	-9.0	-14.0	-56		N/A

1/ Derate linearly 1.2 W/°C for T_C > +25°C; $P_T = (T_{jmax} - T_C)/R_{\theta JC}$

2/ $I_D = ((T_{jmax} - T_C)/((R_{\theta JC})X(r_{DS(on)} \text{ at } T_{jmax})))^{1/2}$

3/ I_{DM} = 4 X I_{D1}; I_{D1} as calculated by footnote 2/.

4/ Unless otherwise noted, electrical characteristics, ratings, and conditions for “U” suffix devices (surface mount) are identical to the corresponding non-”U” suffix devices.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_C = +25°C.

Type 2/	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = -1.0 mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = -1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = -12V $I_D = I_{D2}$		$R_{\theta JC}$ Max	E_{AS} at I_{D1}	I_{AS}
					T_J = 25°C	T_J = 150°C			
	V dc	V dc		µA dc	Ω	Ω	°C/W	mJ	A
		Min	Max						
2N7422	-100	-2.0	-4.0	-25	0.080	0.200	0.83	500	-22
2N7423	-200	-2.0	-4.0	-25	0.315	0.708	0.83	500	-14

1/ Pulsed (see 4.5.1).
2/ Unless otherwise noted, electrical characteristics, ratings, and conditions for “U” suffix devices (surface mount) are identical to the corresponding non-”U” suffix devices.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

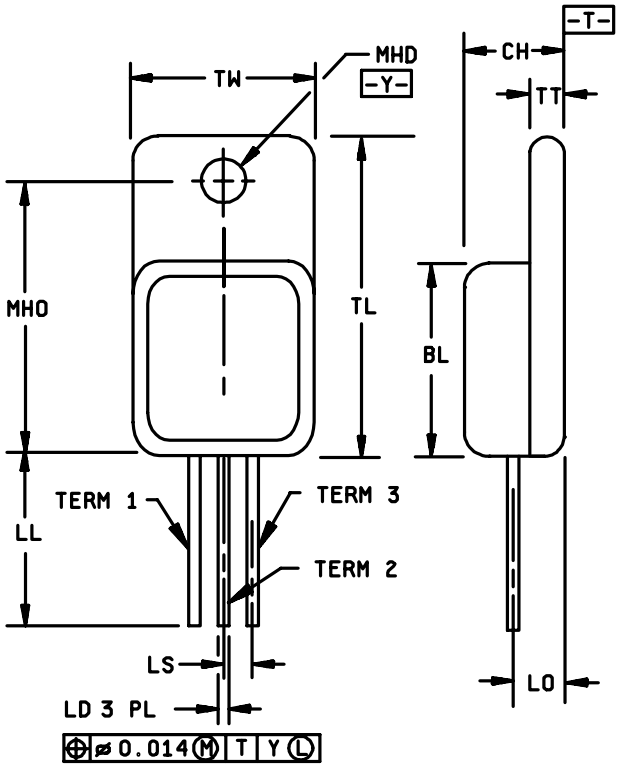
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.535	.545	13.59	13.89
CH	.249	.260	6.32	6.60
LD	.035	.045	0.89	1.14
LL	.510	.570	12.95	14.48
LO	.150 BSC		3.81 BSC	
LS	.150 BSC		3.81 BSC	
MHD	.139	.149	1.53	3.78
MHO	.665	.685	16.89	17.40
TL	.790	.800	20.07	20.32
TT	.040	.050	1.02	1.27
TW	.535	.545	13.59	13.89
Term 1	Drain			
Term 2	Source			
Term 3	Gate			



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Refer to applicable symbol list.
4. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
5. All terminals are isolated from case.

FIGURE 1. Physical dimensions for TO-254AA.

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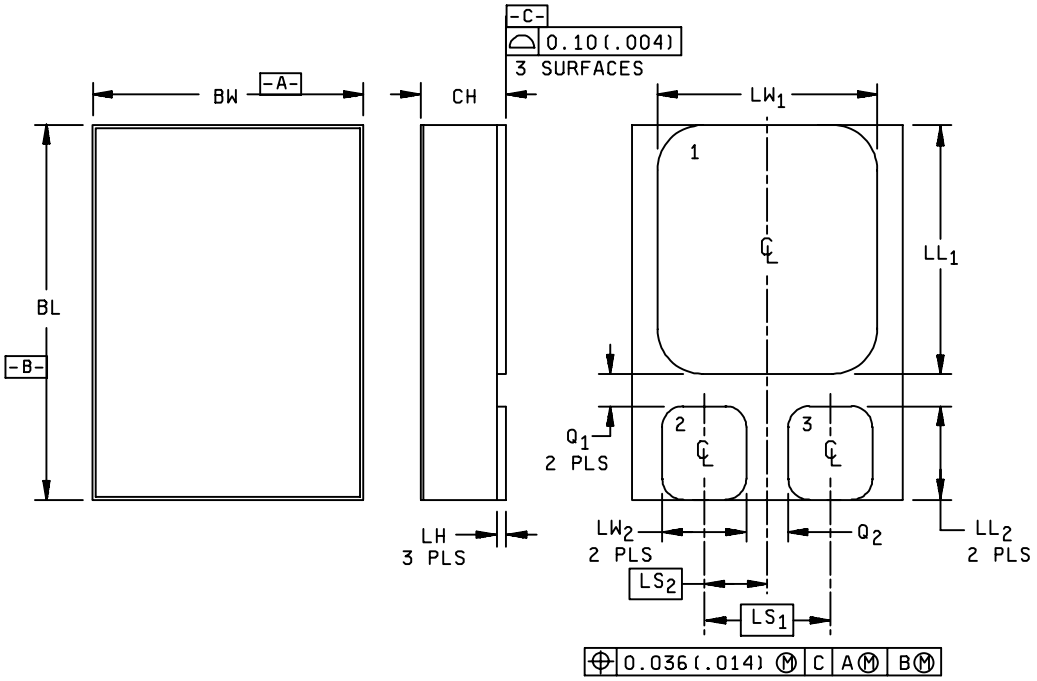


FIGURE 2. Dimensions and configuration of surface mount package outline.

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Dimensions				
Symbol	SMD-1			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.619	.631	15.73	16.02
BW	.444	.456	11.28	11.58
CH	- - -	.142	- - -	3.60
LH	.010	.020	0.26	0.50
LL_1	.409	.421	10.39	10.69
LL_2	.151	.163	3.84	4.14
LS_1	.210 BSC		5.33 BSC	
LS_2	.105 BSC		2.67 BSC	
LW_1	.369	.381	9.38	9.67
LW_2	.134	.146	3.41	3.70
Q_1	.030	- - -	0.76	- - -
Q_2	.035	- - -	0.89	- - -
Term 1	Drain			
Term 2	Gate			
Term 3	Source			

NOTES:
1. Controlling dimension: Inch.
2. Metric equivalents are given for information only.
3. The lid shall be electrically isolated from the drain, gate, and source.
4. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Dimensions and configuration of surface mount package outline - Continued.

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3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1 and 2 herein. Methods used for electrical isolation of the terminals shall employ materials that contain a minimum of 90 percent Al_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets, or ceramic walled packages.

3.3.1 Lead material and finish. Lead material shall be kovar or Alloy 52; a copper core or plated core is permitted. Surface mount terminals shall be copper tungsten. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of terminal finish is desired, it shall be specified in the acquisition document (see 6.2).

3.3.2 Internal construction: Multiple chip construction shall not be permitted to meet the requirements of this specification.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the transistor but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic discharge protection (see 6.2).

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

3.6 Electrical performance characteristics: Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-PRF-19500, and table III herein.

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4.3 Screening (JANS and JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS	JANTXV
1/	Method 3470, E_{AS} test (see 4.5.4)	Method 3470, E_{AS} test (see 4.5.4)
1/	Method 3161, thermal impedance (see 4.5.3)	Method 3161, thermal impedance (see 4.5.3)
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
9 2/	I_{GSS1}, I_{DSS1}, Subgroup 2 of Table I herein	Not applicable
10	MIL-STD-750, method 1042 test condition B	MIL-STD-750, method 1042 test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)}$, $V_{GS(TH)}$ Subgroup 2 of table I herein.
12	MIL-STD-750, method 1042 test condition A	MIL-STD-750, method 1042 test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.

1/ Shall be performed anytime before screen 10.
2/ Shall be performed after E_{AS} test, thermal impedance test, and gate stress test.

4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with figure 4 of MIL-PRF-19500.

4.1 Group A inspection. Group A inspection shall be conducted in accordance with table V of MIL-PRF-19500 and table I herein. Electrical measurements (endpoints) shall be in accordance with table I, group A, subgroup 2 herein.

4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end points) shall be in accordance with table I, group A, subgroup 2 herein.

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4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	1051	Test condition G, 100 cycles
3	2077	SEM
4	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. ton = 30 seconds minimum.
5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C, t = 24 hours minimum, or T_A = +150°C, t = 48 hours minimum.
5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C, t = 120 hours minimum; or T_A = +150°C, t = 240 hours minimum.
5	2037	Bond strength (Al-Au die interconnects only), Test condition A
6	3161	Thermal resistance, see 4.5.2.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
2	1051	Test condition G, 25 cycles.
3	1042	Intermittent operation life, condition D, 2,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. ton = 30 seconds minimum.
5 and 6	----	Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500 and as follows. Electrical measurements (end points) shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
2	2036	Test condition A; weight = 10 pounds; t = 15 seconds (applicable to TO-254AA only).
6	1042	Intermittent operation life, condition D, 6,000 cycles. No heat sink or forced-air cooling on the device shall be permitted during the on cycle. ton = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table VIII of MIL-PRF-19500 and table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal Resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\theta JC}$ = 0.83 °C/W. The following parameters shall apply:

a. Measuring current(I_M).....10 mA.

b. Drain heating current (I_H).....4.0 A (5.0 A for "U" suffix devices) .

c. Heating time (t_H).....Steady-state (see MIL-STD-750, method 3161).

d. Drain-source heating voltage (V_H).....25 V (20 V for "U" suffix devices)

e. Measurement time delay (t_{MD}).....30 to 60 μs.

f. Sample window time (t_{SW}).....10 μs maximum.

4.5.3 Thermal Response ($Z_{\theta JC}$ measurement). The $Z_{\theta JC}$ measurement (or equivalent ΔVSD measurement) shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed figure 2, thermal impedance curves and the group A, subgroup 2 limits) for $Z_{\theta JC}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot, to be plotted on the applicable X and R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition. This procedure may be used in lieu of an in line process monitor.

a. Measuring current (I_M).....10 mA.

b. Drain heating current (I_H).....4 A (5 A for "U" suffix devices).

c. Heating time (t_H).....100 ms (30 ms for "U" suffix devices).

d. Drain-source heating voltage (V_H).....25 V (20 V for "U" suffix devices).

e. Measurement time delay (t_{MD}).....30 - 60 μs.

f. Sample window time (t_{SW}).....10 μs maximum.

4.5.4 Single pulse avalanche energy (E_{AS}).

a. Peak current, $I_{AS} = I_{D1}$

b. Inductance, $L = (2*E_{AS}/(I_{D1})^2)*((V_{BR}-V_{DD})/V_{BR})$ mH minimum

c. Gate to source resistor, R_{GS}: $25 \leq R_{GS} \leq 200$ Ω

d. Supply voltage, V_{DD} = -25 V dc, except V_{DD} = -50 V dc for 2N7423

e. Initial case temperature, T_C = 25° C, -5° C, +10 °C

f. Gate voltage, V_{GS} = -12 V dc

g. Number of pulses to be applied: 1 pulse minimum

4.5.5 Gate stress test.

a. V_{GS} = -24 V, minimum

b. t = 250 μS, minimum

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TABLE I. Group A inspection

Inspection 1/, 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3 (TO-254AA) (surface mount)	$Z_{\theta JC}$	 .60 .45		 °C/W °C/W
Breakdown voltage drain to source	3407	V_{GS} = 0 V, I_D = -1 mA dc, Bias condition C	$V_{(BR)DSS}$			
2N7422				-100		V dc
2N7423				-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -1 mA dc	$V_{GS(TH)1}$	-2.0	-4.0	V dc
Gate current	3411	V_{GS} = -20 V dc, Bias condition C, V_{DS} = 0 V	I_{GSSF1}		-100	nA dc
Gate current	3411	V_{GS} = 20 V dc, Bias condition C, V_{DS} = 0 V	I_{GSSR1}		100	nA dc
Drain current	3413	V_{GS} = 0 V dc, Bias condition C, V_{DS} = 80 percent of rated V_{DS},	I_{DSS1}		-25	µA dc
Static drain to source "ON" state resistance	3421	V_{GS} = -12 V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)1}$			
2N7422					0.080	Ω
2N7423					0.315	Ω
Static drain to source "ON" state resistance	3421	V_{GS} = -12 V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D1}$	$r_{DS(ON)2}$			
2N7422					0.085	Ω
2N7423					0.330	Ω
Forward voltage	4011	V_{GS} = 0 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D1}$	V_{SD}			
2N7422					-3.0	V dc
2N7423					-3.6	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/, 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate current	3411	V_{GS} = -20 V dc, Bias condition C, V_{DS} = 0 V	I_{GSSF2}		-200	nA dc
Gate current	3411	V_{GS} = 20 V dc, Bias condition C, V_{DS} = 0 V	I_{GSSR2}		200	nA dc
Drain current	3413	V_{GS} = 0 V dc, Bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		-0.25	mA dc
Static drain to source "ON"-state resistance	3421	V_{GS} = -12 V dc, condition A, Pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)3}$			
2N7422					0.170	Ω
2N7423					0.669	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -1 mA dc	$V_{GS(TH)2}$	-1.0		V dc
Low temperature operation:		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = -1 mA dc	$V_{GS(TH)3}$		-5.0	V dc
Subgroup 4						
Forward transconductance	3475	ID = ID2, V_{DD} > -15 V dc See 4.5.1	g_{fs}			
2N7422					11.0	S
2N7423					4.0	S
Switching time tests	3472	$I_D = I_{D1}$, V_{GS} = -12 V dc R_G = 2.35 Ω, V_{DD} = 50 percent of rated V_{DS}				
Turn-on delay time			$t_{D(on)}$			
2N7422					40	ns
2N7423					60	ns
Rise time			t_r			
2N7422					170	ns
2N7423					240	ns

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/, 3/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 4- continued						
Turn-off delay time			$t_{D(off)}$			
2N7422					190	ns
2N7423					225	ns
Fall time			t_f			
2N7422					190	ns
2N7423					220	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figures 4 and 5 t_p = 10 ms min. V_{DS} = 80 percent of max. rated V_{DS}				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{G(ON)}$			
2N7422					200	nC
2N7423					200	nC
Gate to source charge			Q_{GS}			
2N7422					35	nC
2N7423					45	nC
Gate to drain charge			Q_{GD}			
2N7422					48	nC
2N7423					85	nC
Reverse recovery time	3473	di/dt = -100 A/µs, $V_{DD} \leq$ -50 V $I_D = I_{D1}$	t_{rr}			
2N7422					300	ns
2N7423					775	ns

1/ For sampling plan, see MIL-PRF-19500.

2/ This test required for the following endpoint measurements only:
Group B Subgroups 2 and 3 (JANTX/JANTXV)
Group B Subgroups 3 and 4 (JANS)

Group C, subgroup 6 (All quality levels)
Group E, subgroup 1 (All quality levels)

3/ Unless otherwise noted, electrical characteristics, ratings, and conditions for “U” suffix devices (surface mount) are identical to the corresponding non-”U” suffix devices.

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TABLE II. Group D inspection.

Inspection 1/, 2/, 3/, 6/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits				Unit
				R and F		R		F 4/		
	Method	Conditions		Min	Max	Min	Max	Min	Max	
Subgroup 1										
Not applicable										
Subgroup 2		T_C = + 25°C								
Steady-state total dose irradiation (VGS bias) 5/	1019	V_{GS} = -12 V; V_{DS} = 0 V								
Steady-state total dose irradiation (VDS bias) 5/	1019	V_{GS} = 0 V; V_{DS} = 80 percent of rated V_{DS} (pre-irradiation)								
End-point electricals										
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = -1 mA; Bias condition C	$V_{(BR)DSS}$							
2N7422				-100		-100		-100		V dc
2N7423				-200		-200		-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = -1 mA	$V_{GS(th)1}$							
2N7422				-2.0	-4.0	-2.0	-4.0	-2.0	-5.0	V dc
2N7423				-2.0	-4.0	-2.0	-4.0	-2.0	-5.0	V dc
Gate current	3411	V_{GS} = -20 V V_{DS} = 0 V Bias condition C	I_{GSSF1}		-100		-100		-100	nA dc
Gate current	3411	V_{GS} = +20 V V_{DS} = 0 V Bias condition C	I_{GSSR1}		100		100		100	nA dc
Drain current	3413	V_{GS} = 0 V V_{DS} = 80 percent of rated VDS (pre-irradiation) Bias condition C	I_{DSS}		-25		-25		-25	µA dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/, 2/, 3/, 6/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits				Unit
				R and F		R		F 4/		
	Method	Conditions		Min	Max	Min	Max	Min	Max	
Subgroup 2 - Continued		T_C = + 25°C								
Static drain to source on-state voltage	3405	V_{GS} = -12 V $I_D = I_{D2}$ Condition A pulsed (see 4.5.1)	$V_{DS(on)}$							
2N7422					-1.12		-1.12		-1.12	V dc
2N7423					-2.835		-2.835		-2.835	V dc
Forward voltage source drain diode	4011	V_{GS} = 0 V $I_D = I_{D1}$ Bias condition C	V_{SD}							
2N7422					-3.0		-3.0		-3.0	V dc
2N7423					-3.6		-3.6		-3.6	V dc

1/ For sampling plan see MIL-PRF-19500

2/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other detail specification utilizing the same die design.

3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in it's qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.

4/ The "F" designation represents devices which pass endpoints at both 100K Rads and 300K Rads (Si).

5/ Separate samples shall be pulled for each bias.

6/ Unless otherwise noted, electrical characteristics, ratings, and conditions for "U" suffix devices (surface mount) are identical to the corresponding non-"U" suffix devices.

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TABLE III. Group E inspection (all quality levels) - For qualification only

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			12 devices c = 0
Temperature cycling	1051	Test condition G, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 1/			12 devices c = 0
Steady-state gate bias	1042	Test condition B; 1,000 hours	
Electrical measurements		See Table I, group A, subgroup 2	
Steady state reverse bias	1042	Test condition A; 1,000 hours	
Electrical measurements		See Table I, group A, subgroup 2	
Subgroup 3			
Not applicable			
Subgroup 4			12 devices c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 0.83°C/W maximum. See 4.5.2	
Subgroup 5			
Not applicable			

1/ A separate sample may be pulled for each test condition.

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2N7422, 2N7422U, 2N7423, and 2N7423U

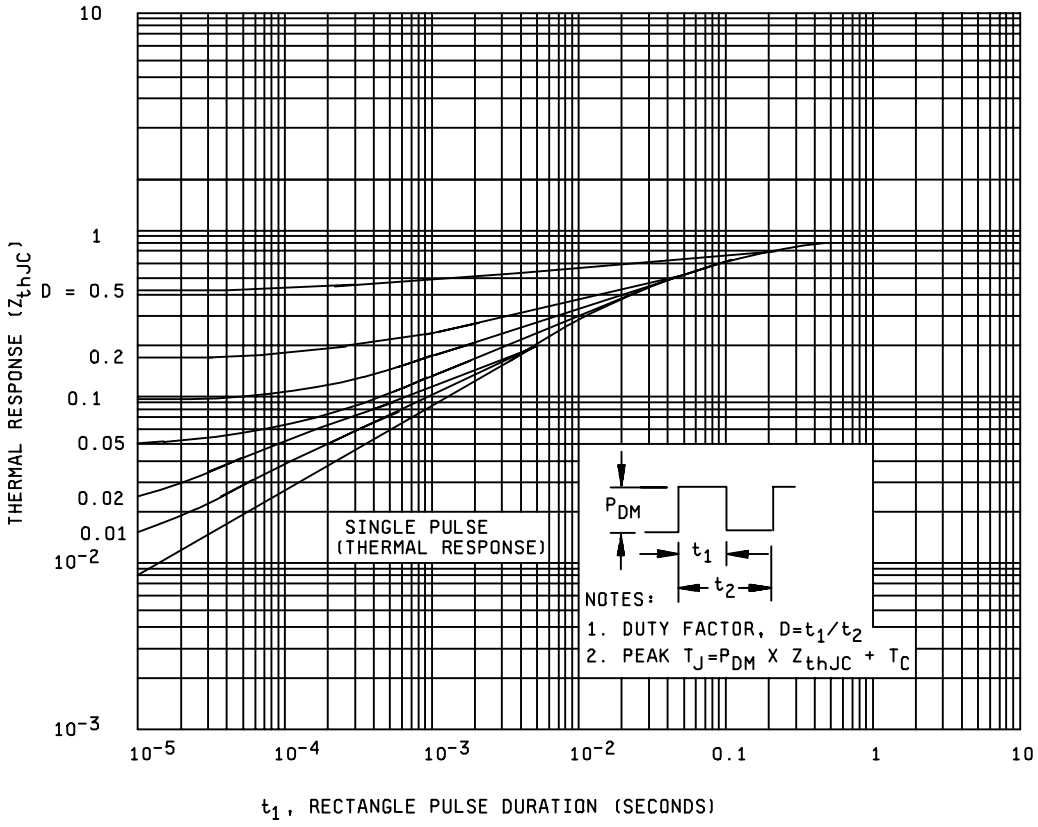


FIGURE 3. Thermal response curve

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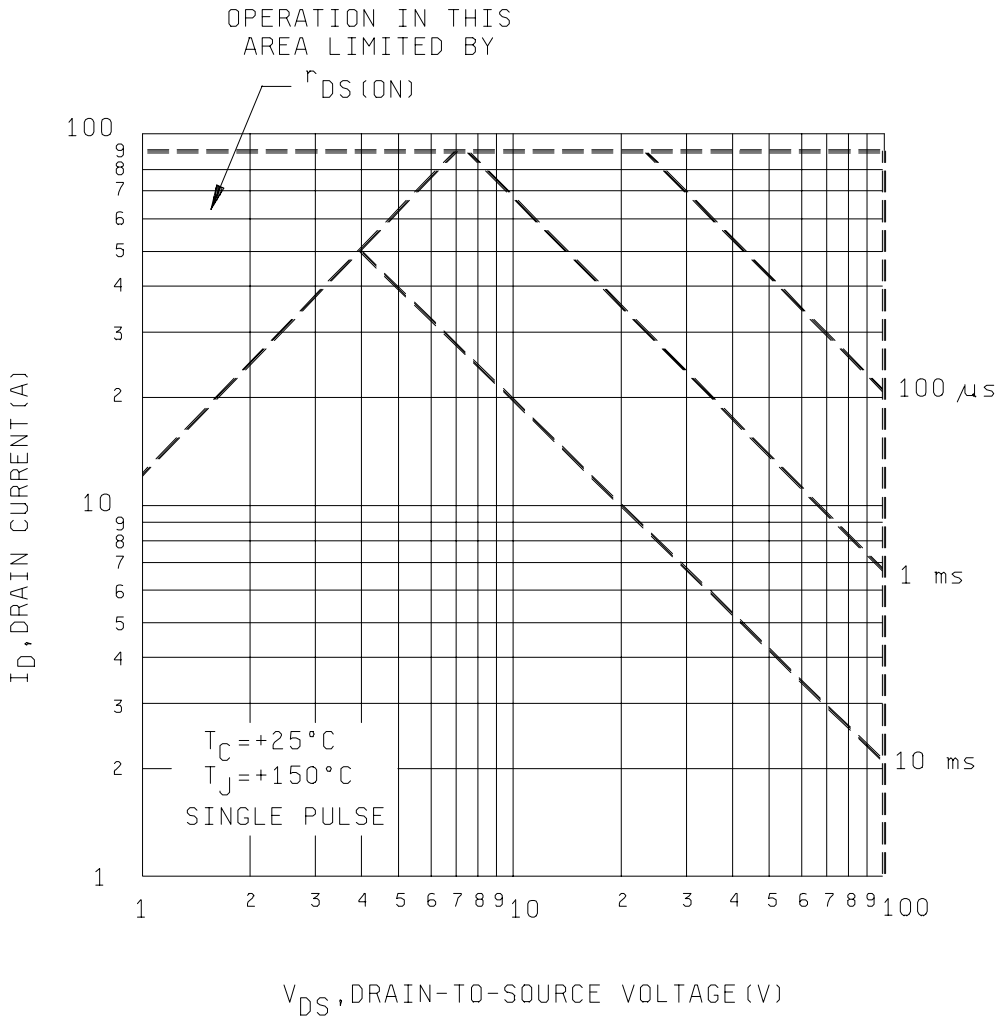


FIGURE 4. Safe operating area graph (2N7422, 2N7422U).

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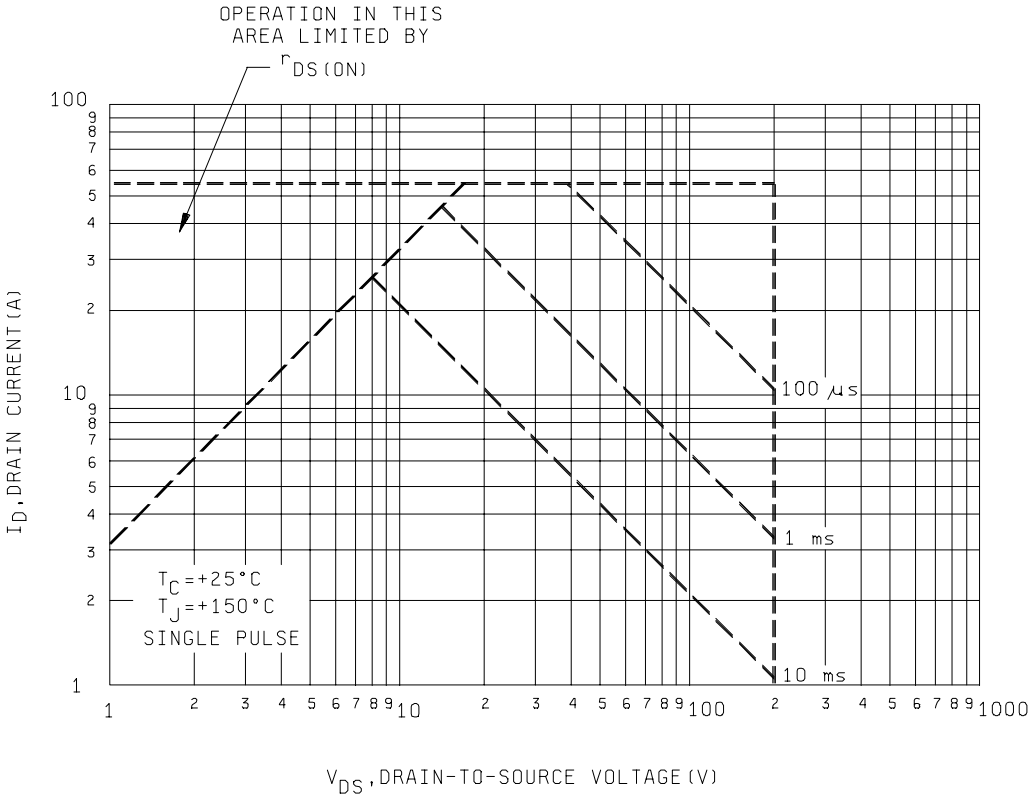


FIGURE 5. Safe operating area graph (2N7423, 2N7423U).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129 and MIL-PRF-19500.

6. NOTES

This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

6.4 Cross-reference list. The following table shows the generic P/N and its associated military P/N (without JAN and RHA prefix).

Commercial types 1/		
Preferred types	TO-254AA	"U"
2N7422	IRHM9_150	IRHN9_150
2N7423	IRHM9_250	IRHN9_250

1/ IRH9_: 100 K Rad (Si)
IRH93: 300 K Rad (Si)

Custodians:
Army - CR
Air Force - 17
Navy - EC
NASA - NA

Preparing activity:
DLA - CC

(Project 5961- 2007)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/662	**2. DOCUMENT DATE** 29 May 1998

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, FIELD EFFECT RADIATION HARDENED(TOTAL DOSE ONLY), P-CHANNEL SILICON TYPES 2N7422, 2N7422U, 2N7423, AND 2N7423U JANTXVR AND F AND JANSR AND F

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) Commercial DSN FAX EMAIL	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. Point of Contact Alan Barone	b. TELEPHONE Commercial 614-692-0510 DSN 850-0510 FAX 614-692-6939 EMAIL alan_barone@dscc.dla.mil
c. ADDRESS Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

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